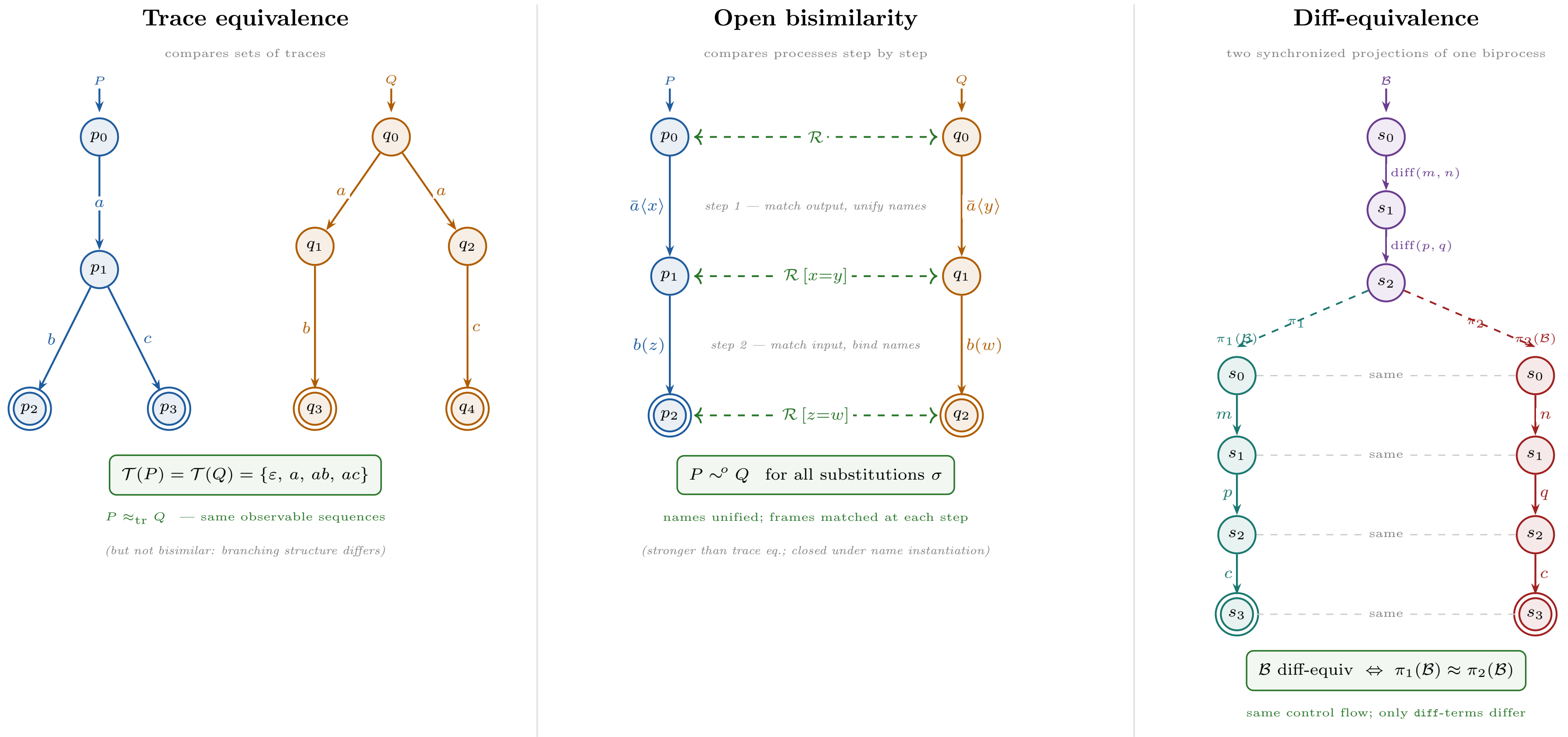

Trace equivalence
compares sets of traces
P
Q
p0
a
p1
b
c
p2
p3
q0
a
a
q1
q2
b
c
q3
q4
$\mathcal{T}(P) = \mathcal{T}(Q) = \{\varepsilon,\ a,\ ab,\ ac\}$
$P \approx_{\mathrm{tr}} Q$ — same observable sequences
(but not bisimilar: branching structure differs)
Open bisimilarity
compares processes step by step
P
Q
p0
$\mathcal{R}$
q0
$\bar{a}\langle x\rangle$
step 1 — match output, unify names
$\bar{a}\langle y\rangle$
p1
$\mathcal{R}\,[x{=}y]$
q1
$b(z)$
step 2 — match input, bind names
$b(w)$
p2
$\mathcal{R}\,[z{=}w]$
q2
$P \sim^{o} Q$ for all substitutions σ
names unified; frames matched at each step
(stronger than trace eq.; closed under name instantiation)
Diff-equivalence
two synchronized projections of one biprocess
$\mathcal{B}$
s0
diff(m, n)
s1
diff(p, q)
s2
π_1
π_2
$\pi_1(\mathcal{B})$
$\pi_2(\mathcal{B})$
s0
same
s0
m
n
s1
same
s1
p
q
s2
same
s2
c
c
s3
same
s3
$\mathcal{B}$ diff-equiv $\Leftrightarrow$ $\pi_1(\mathcal{B}) \approx \pi_2(\mathcal{B})$
same control flow; only diff-terms differ